

Saw-Stage Early Warning for Hidden Downstream Quality Defects

OCTWAVE 3.0 | Smart Manufacturing & Automation | Technical report

Executive summary. A retrospective Random Forest prototype uses completed Saw signals to flag workpieces for inspection before Milling. On a selected fixed benchmark it detected **20/30 target defects (66.67%)**, with **1/45 non-target false alarms (2.22%)**. The experiment supports an inspection opportunity within represented production days; new-day generalization and operational savings remain unproven.

1. Problem and manufacturing context

Acceptable weight does not ensure acceptable geometry or surface quality. In the notebook, all 99 documented Saw misalignments passed reconstructed weight QC; 81 subsequently failed Milling QC, 16 passed and 2 had no Milling record [5]. Quality engineers and production operators therefore need a way to identify risky workpieces before further machining resources are consumed.

Research question: Can a completed Saw operation identify weight-QC-passing workpieces that later fail Milling QC in the documented Saw-misalignment fault class? The objective is high target capture with a low additional-inspection burden, not automatic scrapping or prediction of every downstream failure.

2. What the dataset contains

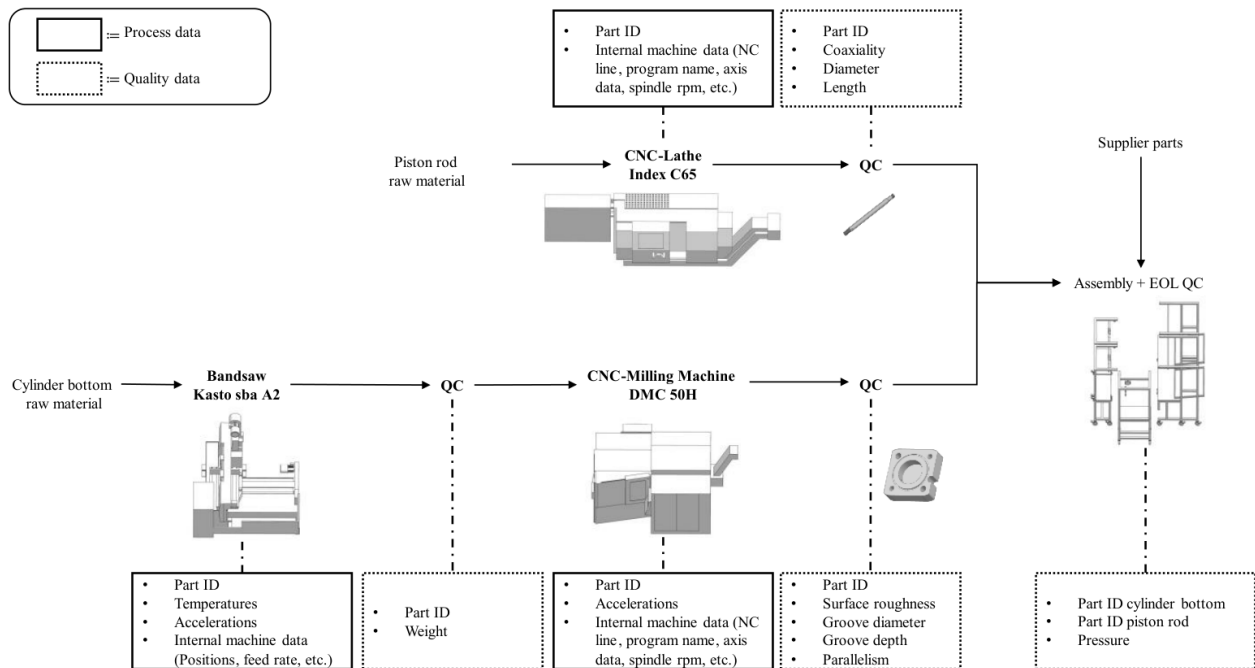


Figure 1. Full manufacturing chain and recorded data. Reproduced from Jourdan et al., Fig. 1 [7], without changing the figure content; scaled for layout. The figure includes both component branches. This project models only the cylinder-bottom Saw branch. Figure licence: CC BY-NC-ND 4.0 as printed in the deposited manuscript; attribution details on page 6.

CiP-DMD documents a pneumatic-cylinder learning factory at TU Darmstadt. The publication describes 847 assembled cylinders; that is a different unit and population from this notebook's 984 prepared Saw records. QR-based part IDs connect signals and quality measurements across stages [1,2,7]. The source mechanism is a deliberately short blank affecting subsequent face milling; the notebook calls its annotation Saw misalignment.

Data understanding and methodology

3. Data selection and target construction

Available data	Use in this study
Saw process time series	Active-cut power, energy, feed, position and vibration features are predictors.
Saw weight, Milling measurements	Historical QC rule and target construction only; never model inputs. Milling checks roughness, parallelism, groove depth and diameter.
Fault annotations, part IDs, timestamps	Define target class, link records and form validation groups. Dates and IDs are excluded from predictors.
Turning, assembly and additional Milling signals	Dataset context; not inputs to the final RF. Assembly pressure pass/fail is not this prediction target.

Target = 1: documented Saw misalignment AND Saw weight pass AND known Milling QC failure. Target = 0 covers the other known outcomes within the selected population, including class-0 products that fail Milling. Unknown outcomes are excluded. This is supervised prediction of a specific quality escape, not unsupervised anomaly detection.

Population step	Products	Interpretation
Prepared Saw / class 0 or 1	984 / 933	Other documented fault classes excluded from the initial task.
Usable active-cut / mixed-class days	931 / 474	Six days retained with both original task labels.
Initial development / fixed test	383 / 91	Blocks of eight consecutive products.
Known Milling outcomes only	404	70 excluded for incomplete required outcomes.
Final development / benchmark	329 / 75	33 / 30 target positives; 296 / 45 negatives.

4. Feature engineering, fitting and threshold

Use samples with **bCutActive** > 0.5 and at least five active samples. Extract variation and shape statistics: standard deviation, IQR, range, slope, skewness, kurtosis and absolute changes. From 168 allowed candidates, 129 remain in the final fit. Per training fold, remove >20% missing, constant/near-constant and highly correlated features ($|r| > 0.95$), and impute training medians. Apply those learned transformations to validation/test data.

RF configuration: 500 trees, depth 6, minimum leaf size 2, balanced class weights, seed 42. The shallow depth and leaf constraint limit complexity for a small positive class; no architecture search or optimality claim is made. No pretrained weights are used.

Validation: GroupShuffleSplit (seed 42) selects the first split satisfying the implemented size/class/day-coverage conditions, out of at most 500 candidates. Both partitions contain every selected day. Five-fold StratifiedGroupKFold on development blocks (seed 123) supplies out-of-fold scores. Search thresholds 0.05-0.80, step 0.025; maximize capture subject to development false alarms $\leq 5\%$, then precision and threshold. The selected threshold is **0.25**; final fitting uses all 329 development products.

Leakage boundary: future quality and fault labels define y, not X. Cleaning is training-only in the final pipeline. However, the dataset was explored before the split and the benchmark was later reused: this is a fixed benchmark, not an untouched confirmation of the entire research process.

Model quality and interpretation

5. Fixed-benchmark performance

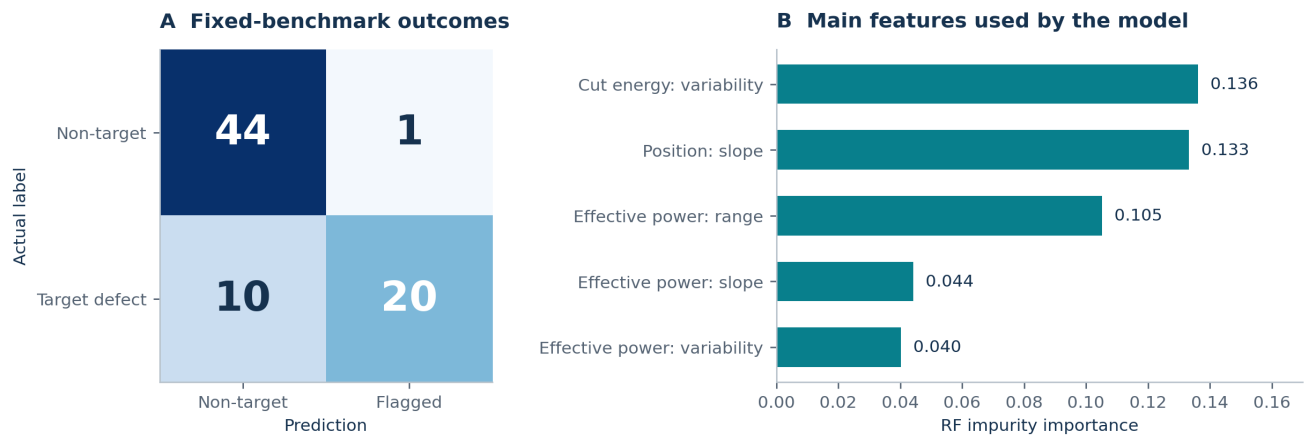


Figure 2. Final business-target RF outcomes and five largest impurity importances, redrawn from saved notebook results [5]. Feature importance indicates model reliance, not a causal mechanism. No raw predictions were simulated to create these figures.

Measure	Random Forest	TCN
Target capture	20/30 = 66.67%	14/30 = 46.67%
False-alarm rate	1/45 = 2.22%	0/45 = 0.00%
Precision / F1	95.24% / 0.784	100.00% / 0.636
Balanced accuracy	0.822	0.733
ROC-AUC / average precision	0.914 / 0.897	0.762 / 0.812
Development-selected threshold	0.25	0.875

RF caught more target defects; TCN made fewer false alarms in this sample. Both used the same 75 benchmark products but different representations, fitting procedures and operating thresholds. These are pipeline comparisons, not an architecture-only test at identical false-alarm rates. The existing weight rule flagged none of the 30 target defects; its value is as an operational reference, not a comprehensive comparison with geometric inspection.

TCN training: 12 channels resampled to 128 points; a 32-channel stem, four residual dilated blocks (1, 2, 4, 8), average/max pooling and a binary head. Weighted BCEWithLogitsLoss follows the forward pass; backpropagation and AdamW (learning rate 0.001, weight decay 0.0001) update parameters. Training-fold normalization and validation early stopping are used; benchmark predictions average fold models. The stored run used CPU [5].

Development evidence: RF out-of-fold scores at the selected threshold caught 30/33 positives with 10/296 false alarms. MiniROCKET caught 23/33 with 12/296 false alarms and was evaluated only on development. Threshold-selection scores are not independent test results. Earlier apparent capture near 95% included training products and is excluded from final claims.

6. What performance does and does not mean

The RF misses ten target defects. Its one false alarm is a class-0 product passing Milling QC. Final-RF errors still need a per-day/block audit. Benchmark target prevalence is 40%, versus 10.03% in development, so 95.24% precision is not a factory expectation. Average precision is the sklearn score labelled PR-AUC in the notebook; it is not necessarily trapezoidal area. No cluster-aware uncertainty intervals or score calibration are established.

Early detection: promise and stopping decision

Why pause after good results? The timing result is worth retaining, but the follow-up analysis could not separate fault-related signal from production-period effects. We paused the validated within-cut claim and further model search; we did not conclude that early detection is impossible.

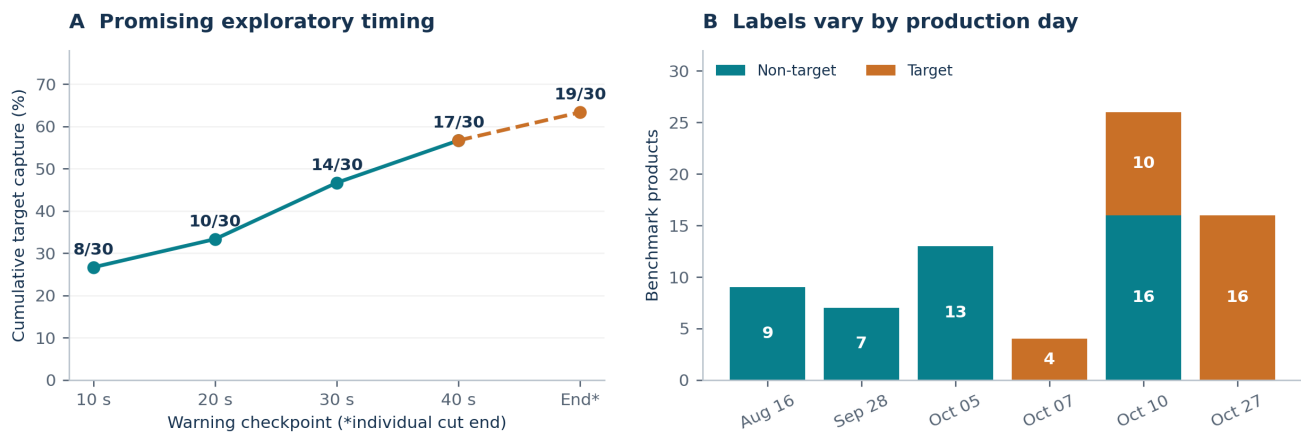


Figure 3. Exploratory progressive capture and the same benchmark's day composition [5]. The End checkpoint is categorical because cut durations differ; it is not plotted as a fixed 52 seconds. Single-class days show why aggregate performance can be misleading.

7. Promising result, correctly separated

The progressive policy evaluates 10, 20, 30 and 40 seconds, then cut end, with development-selected thresholds controlling its overall alert rate. It flagged **19/30** targets and **2/45** non-targets. Of the targets, **17/30 (56.67%)** were detected during cutting, two only at the end, and eleven were missed. Among the 19 detected targets, median first warning was 20 seconds. Among the 17 true within-cut detections, median lead time was 54 seconds before cut completion; these timing statistics exclude missed cases.

Diagnostic	Observed result	Meaning
Same-day 30-second score pairs	10 pairs; only 2 defect-score wins	No supportive pairwise discrimination in this small subset.
Matching quality	0 pairs within 60 min; 1 same block	A strict matched-condition test was not feasible.
Cut duration	Early-detected median 74 s; missed 49 s	Warning success co-varies with operating period/duration.

The important distinction: The standalone 30-second classifier caught 16/30 at threshold 0.25. The progressive policy had caught 14/30 by 30 seconds under its different thresholds and accumulated alerts. The paired diagnostic examines 30-second scores, not the complete progressive policy; it therefore raises a concern rather than conclusively invalidating every checkpoint.

An earlier strict-matched, full-cut misalignment experiment also performed poorly (pooled ROC-AUC 0.458, F1 0.383). That was a different target/model from the final business RF. Neither this result nor the early diagnostic proves the final RF is robust; neither proves the early idea cannot work. More fitting on the repeatedly inspected benchmark would not resolve this evidence gap.

Retained scope: full-Saw prediction is early relative to downstream Milling. Reliable detection in the first seconds of cutting remains an exploratory extension requiring independent, time-valid validation.

Industrial use, contribution and next steps

8. Research context and differentiation

CiP-DMD was released to support linked manufacturing analysis [1,7]. SSMSPC uses self-supervised representations and statistical monitoring for anomaly detection/localization, including a CNC-Milling subset of CiP-DMD [3]. Its target, sensor location and protocol differ from this study, so its published scores are not benchmark comparators. Here, the contribution is a specific **Saw-QC-pass / Milling-fail target** and an earlier inspection opportunity using Saw signals. This is application framing, not a new RF algorithm or a claim to be first. Manufacturing reliability research also motivates drift-aware evaluation and monitoring [4].

9. Proposed workflow and business value

At Saw completion, link signals to the workpiece ID, apply saved preprocessing and score the product. A weight-QC-passing workpiece above the threshold is held for operator inspection before Milling; a low-score workpiece follows normal production and QC. Missing or invalid sensors fall back to existing procedures. No automatic scrap or unrestricted release decision is proposed.

A deployment would version the model, features and threshold; log scores and actions; monitor missingness, input/score drift and delayed-label capture/false alarms; and support rollback. These are proposed MLOps functions, not implemented deployment evidence. Potential net value is avoided downstream processing cost minus additional inspection, holding and operating cost. No measured savings, throughput benefit or inference latency is claimed.

10. Principal limitations

Evidence boundary	Implication
Selected mixed-class days; same days in both splits	Operating conditions are not experimentally controlled. New-day, new-machine and factory transfer are unproven.
Small sample; 33 training positives; repeated benchmark use	Rates are uncertain and may reflect research-selection effects. Keep final claims retrospective.
Narrow labels, induced faults and incomplete QC	Results do not cover every fault origin, natural production prevalence or final assembly/customer quality.
Unmeasured intervention and incomplete reproducibility audit	Operator effectiveness, costs, latency, exact source/dependency versions and clean-run execution still need verification.

11. Further development: make early warning testable

First - establish independent evidence. Predefine the target, segmentation, feature rules, checkpoint policy and thresholds. Evaluate future production periods with both classes under comparable settings, enough close controls and temporal gaps between splits. Audit final full-Saw RF errors before attributing the signal to faults.

Then - revisit within-cut detection. Train only on data available up to each checkpoint, with causal segmentation and no completed-cut normalization or duration information. Select thresholds on development only and control the false-alarm rate of the entire repeated-alert policy. Report capture, coverage, missed cases, first-warning time and lead time jointly, including day/block uncertainty and comparison with the full-stage baseline.

Finally - test practical value. Investigate stable features or representations only against a diagnosed weakness; calibrate scores, compare simple baselines and direct inspection options, measure costs/latency, and run an operator-reviewed shadow pilot. New data collection belongs after the challenge or under explicit organizer permission.

References and source provenance

References and appendices are outside the five-page main report, following the booklet’s recommended limit [6].

Research and dataset sources

- [1] N. Jourdan, T. Biegel, B. Bretones Cassoli and J. Metternich. *A new benchmark dataset for machine learning applications in discrete manufacturing: CiP-DMD*. Procedia CIRP 126 (2024), 423-428. [Published paper](#) / [DOI](#).
- [2] HannesJoern. *interq_data_acquisition_public*, public dataset/code repository and README. [Repository and data structure](#). Accessed 11 September 2026. The exact commit used by the original notebook was not recorded in its saved output.
- [3] T. Biegel, P. Helm, N. Jourdan and J. Metternich. *SSMSPC: self-supervised multivariate statistical in-process control in discrete manufacturing processes*. Journal of Intelligent Manufacturing 35 (2024), 2671-2698; online 2023. [Paper](#) / [DOI](#).
- [4] N. Jourdan, S. Sen, E. J. Husom, E. Garcia-Ceja, T. Biegel and J. Metternich. *On The Reliability Of Machine Learning Applications In Manufacturing Environments*. NeurIPS Workshop on Distribution Shifts (2021), arXiv:2112.06986. [Open paper](#).
- [5] Project notebook, *test-v7-0.ipynb*, supplied saved code and outputs. Primary evidence: FINAL BUSINESS-TARGET MODEL; DIRECT RAW TIME-SERIES MODEL; DEVELOPMENT-ONLY MINIROCKET EXPERIMENT; REAL-TIME FIXED-TIME EARLY DEFECT DETECTION; VERIFY EARLY-WARNING TIMING; FINAL MATCHED-CONDITION CHECK. The original 92 cells are retained unchanged.
- [6] IEEE IAS, University of Moratuwa Student Branch Chapter. *OctWave 3.0 Final Round Delegate Booklet*. Supplied competition document, especially sections 10-18. Any later official instructions supersede the booklet.
- [7] Jourdan et al. *CiP-DMD*, author-deposited manuscript, 9 October 2023. [Open manuscript and download](#); DOI [10.5281/zenodo.8420132](#). Used to read the process description and reproduce Figure 1. Final publication details are given in [1].

Dataset access and figure rights

The manuscript directs researchers to [the CiP-DMD dataset landing page](#) and states research availability. The notebook uses [2] and a public author-hosted share. The raw dataset’s complete licence/usage terms still need to be retained with the competition submission; the article licence should not be assumed to license every separately hosted data file.

Figure 1 credits Jourdan et al. and reproduces the complete figure content without modification (page surroundings omitted; proportional scaling only). The deposited PDF prints [CC BY-NC-ND 4.0](#); Zenodo metadata instead lists CC BY 4.0. This report follows the more restrictive printed figure attribution and is prepared for noncommercial academic competition use. All result charts are newly drawn from the project’s saved numerical outputs.

Reproducibility and AI assistance

The report was checked against saved outputs; models were not rerun during report production. A fresh Kaggle run, dependency/version capture and input access checks remain required. ChatGPT assisted with problem refinement, code/debugging, interpretation, figures and documentation. Participants must review, test, understand and defend all submitted work.

Appendix A - technical audit and claim limits

This appendix consolidates limitations that should remain visible in the report and Q&A. Further detail and unsuccessful experiments are preserved in the notebook.

Topic	Specific limitation / action
Grouping and confounding	Groups contain eight products, not verified homogeneous process conditions. Adjacent groups may be near in time and no temporal gap is enforced. Metadata-associated signals can persist after IDs/dates are excluded.
Research selection	The final cleaning/fitting excludes benchmark rows, but prior exploration, target evolution and repeated comparisons expose benchmark outcomes. Do not call it an untouched end-to-end confirmation.
Missingness and population	984 prepared records reduce to 404 business-labelled records. Other fault classes, single-class days and incomplete QC are excluded. Missing outcomes may be nonrandom; performance on excluded populations is unknown.
Target versus cause	Only two misaligned Milling-pass products remain in the final population. Separation of failure risk from misalignment itself is weakly supported. Other-origin failures are target-negative, and no multiclass root-cause locator is trained.
QC and linkage	Weight limits are reconstructed from the full source weight distribution, not learned prospectively on development. Validate raw-unit Milling bounds, duplicate resolution and part-ID links against released QC specifications.
Source versus notebook	The paper reports 847 assembled cylinders and a November 2022-June 2023 collection period. Notebook Saw timestamps include August-October 2022. Also, published and prepared fault counts differ outside class 1. Reconcile release, component and stage coverage; do not silently equate the populations.
Small sample and prevalence	Only 30 positives/45 negatives in the benchmark; a single extra false alarm changes its rate by 2.22 points. Target prevalence is 40% versus 10.03% in development. Calibration and cluster-aware intervals are not supplied.
Historical versus final pipeline	Some early cleaning preceded cross-validation and some early predictions included training records. Those outputs are preserved as exploration, not equivalent evidence to the final RF.
Broader failure analysis	50/63 target flags and 0/94 other-origin Milling-failure flags combine development out-of-fold and benchmark predictions. They are not an independent 94-product validation or proof of causal specificity.
Early warning evaluation	Ten same-day pairs are nonrandom and 76-238 minutes apart. One-sided binomial $p=0.989$ and Wilcoxon $p=0.997$ for defect-higher scores offer no support in that subset; pairing does not establish exchangeability or explain the entire policy.
Comparators and explanations	TCN thresholds and representations differ from RF; zero observed false alarms is not a zero underlying rate. MiniROCKET is development-only. Impurity importance is noncausal. Add simple baselines and matched operating-point comparisons later.
Operational and documentation gaps	No intervention trial, quantified savings, measured latency or implemented MLOps. Verify sensor availability, calibrated thresholds, versioned preprocessing/model artifacts, licences and a clean-kernel run. Keep all existing QC.

Review gate for resuming early detection: enough independent positive/negative production periods; predefined causal checkpoints; development-only tuning; an untouched evaluation set; comparable conditions; overall policy false-alarm control; and lead-time distributions that include coverage and missed-case accounting. Until these gates are met, use the result as a research direction rather than a validated real-time capability.